

Agenda

MLE at Google + Gmail

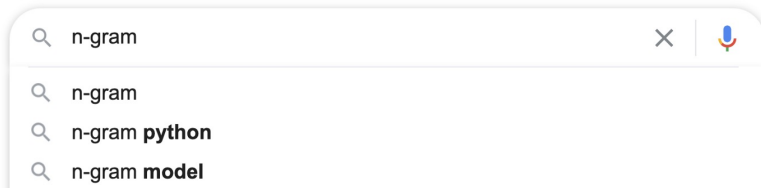
"Hi, hope you are doing —"

well
good
great }

"Maths behind
finding the right
word."

Next Word Prediction

Google



Language Modelling

1. Sentence Probability
2. Next word Probability

↙
previous word / sentence

Probability

$w_1 w_2 w_3 w_4 \dots w_n$

$$\hookrightarrow P(w_t | w_1, w_2, \dots, w_{t-1})$$

Count for every possible history

$w_1 w_2 w_3 \dots w_{t-1}$
X

└ _____

_____ . In the end
it was all _____

Curse of History

S — The water is transparent → $P(\text{transparent} | \text{The water is})$

└ $P(\text{the}) \times P(\text{water}) \times P(\text{is}) \times P(\text{transparent})$

⇒ $P(\text{The} | \text{_____}) \times P(\text{water} | \text{The. _____}) \times P(\text{is} | \text{water, the})$

Markov Assumption & n-gram Approximation

The next word depends on last few words & not the entire history

$$P(w_t | w_1, w_2, \dots, w_{t-1}) \approx P(w_t | w_{t-n+1}, \dots, w_{t-1})$$

n-gram

unigram	→	0 <u>historical</u> words	1 current
bigram	→	1 <u>historical</u> word	1 current
trigram	→	2 <u>historical</u> words	1 current

"I have a dream"

Unigram [I] [have] [a] [dream]

Bigram [I have] [have a] [a dream]

Trigram [I have a] [have a dream]

$$P(\text{I have a dream})$$

$$= P(I) \times P(\text{have}) \times P(a) \times P(\text{dream})$$

Unigram

Bigram

$$P(I \text{ have a dream}) = P(I / \langle \text{start} \rangle) \times P(\text{have} / I) \times P(a / \text{have}) \times P(\text{dream} / \text{have})$$

Tri-gram

$$P(I \text{ have a dream}) = P(I / \langle \text{start} \rangle, \langle \text{start} \rangle) \times P(\text{have} / \langle \text{start} \rangle, I) \times P(a / \text{have}, I) \times P(\text{dream} / a, \text{have})$$

$\langle \text{start} \rangle I \text{ have a dream} \langle \text{end} \rangle$
 $\langle \text{start} \rangle I \text{ ~~have~~ a dog} \langle \text{end} \rangle$
am

Counts:

$$C(\langle \text{start} \rangle, I) = 2$$

$$C(I, \text{have}) = 2$$

$$C(\text{have}, a) = 2$$

$$C(a, \text{dream}) = 1$$

$$C(a, \text{dog}) = 1$$

Current $\downarrow$ Historical $\nearrow$
 $P(\text{dream} / a) = \frac{1}{2}$
bigram

$$\begin{cases}
 P(a / I, \text{have}) = \\
 P(\text{dream} / a, \text{have}) = 0.5 \\
 P(\text{dog} / a, \text{am}) = 1
 \end{cases}$$

$$P(\text{dog} / a) = 1/2$$

$$P(I / \langle \text{start} \rangle) = 2/2 = 1$$

$$P(a / \text{have}) = 2/2 = 1$$

I have a

I have a dream
dog

```
for sentence in corpus:
    for (w1, w2) in bigrams(sentence):
        counts[w1][w2] += 1

for w1 in counts:
    total = sum(counts[w1].values())
    for w2 in counts[w1]:
        probs[w1][w2] = counts[w1][w2] / total
```

Unigram | Bi-gram | tri-gram

more probability of grammatical sentences

text size = 10,000

$$P(I) \times P(a) \times \underbrace{P(-)} \times P()$$

Rare word $\rightarrow \approx 0$

$$0.6 \times 0.3 \times 0.2 \times \dots \approx 0.0003$$

tending to 0

Language Model

→ Log likelihood
→ Cross-entropy
→ Perplexity



$$0.3 \times 0.4 \times 0.8 \dots \rightarrow 0$$

$$P(w_1, w_2, \dots, w_T) = \prod P(w_t | \text{history})$$

$$\log P(w_1, w_2, \dots, w_T) = \underline{\sum \log P(w_t | \text{history})}$$

$$\log(a \times b) = \log a + \log b$$

$$\begin{array}{c} \downarrow \\ \text{Cross-entropy} \\ \text{(H)} \end{array} = - \frac{1}{\text{(T)}} \sum \log P(w_t | \text{history})$$

Average -ve log probability
↓ H Model assigns higher probability to Actual words

↓

✓ Perplexity $\Rightarrow 2^{(H)}$

$2^H \rightarrow$ If H is low then 2^H is also low

Better Model

Model A $\rightarrow$ 40
Model B $\rightarrow$ 80 } Same test set

⇓ A word that was never there on a training set

⇓ Out of Vocabulary word $\rightarrow$

Sorry
for

Zero probability problem

<UNK> Tokens

Count $\rightarrow$ 0

$\rightarrow P(\) \rightarrow 0 \rightarrow \underline{0} = 0$

$$[\text{cross-entropy}] \Rightarrow -\log(0) \Rightarrow \underline{\underline{-\infty}}$$

$$\text{Perplexity} \rightarrow 2^{-\infty}$$

Unseen combination, word



$$\text{count} = 4-5$$

$$P(\text{ }) \rightarrow \underline{\underline{Per}}$$

Laplace Smoothing

tri-grams

have a dream > corpus
have a dog

→ have a car ✓ test

$$P(\text{Car} | \text{have, a}) = 0$$

=

$$\frac{(+1) \checkmark}{\cancel{(+1)} \checkmark \quad \text{no. of words in vocab} \checkmark}$$

$$C(a, \text{dream}) = 1 + 1$$

$$C(a, \text{dog}) = 1 + 1$$

$$C(a, \text{car}) = 0 + 1 \text{ (never seen)}$$

$$C(a) = 2 + 1$$

$$P(\text{dream} | a) = \frac{1}{2}$$

$$P(\text{dog} | a) = \frac{1}{2}$$

$$P(\text{car} | a)$$

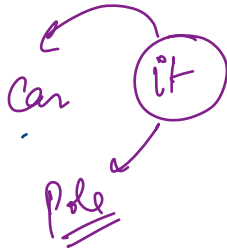
$$= \frac{0 + 1}{\boxed{2} + \textcircled{3}} = \frac{1}{5}$$

Actual when a was there (in first vocab)

a car
a dog
a even

Ambiguity in NLP

- ① Lexical Ambiguity
- ② Syntactic Ambiguity
- ③ Semantic Ambiguity → "Attention is you need".



"first principle code"

"perplexity" $\leftarrow$ $\equiv$
 growing
 by